

PROBLEM SHEET 4

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The following questions are about the simply-typed λ -calculus (STLC).

1. (*) Prove progress and preservation for the **constants-and-functions fragment** of the STLC.

The constants-and-function fragment of the STLC is an extension to the language of numbers and strings: we reached it by *adding* the rules for function types. Thus, to establish these theorems **you only need to show them for the new function rules**, as last week's proofs cover the rest! (We will ignore sums and products in this question!)

Do this in steps:

1. Extend the key lemmata (you may assume weakening, but you can also prove it if you feel like it):
 - (a) Inversion
 - (b) Substitution
 - (c) Canonical forms
2. Prove preservation
3. Prove progress

Solution:

Claim 1 (Inversion). Suppose $\Gamma \vdash e : \tau$.

- If $e = \lambda x : \sigma. e'$ then $\tau = \sigma \rightarrow \tau'$ for some type τ' , and $\Gamma, x : \sigma \vdash e' : \tau'$.
- If $e = e_1(e_2)$ then there exists a type τ_2 such that $\Gamma \vdash e_1 : \tau_2 \rightarrow \tau$ and $\Gamma \vdash e_2 : \tau_2$.

The proof is by inspection.

Claim 2 (Substitution). If $\Gamma \vdash e : \tau$ and $\Gamma, x : \tau \vdash u : \sigma$ then $\Gamma \vdash u[e/x] : \sigma$.

Proof. By induction on the derivation of $\Gamma, x : \tau \vdash u : \sigma$.

Case(LAM). Suppose the derivation is of the form

$$\frac{\displaystyle \frac{\vdots}{\Gamma, x : \tau, y : \sigma_1 \vdash u : \sigma_2}}{\Gamma, x : \tau \vdash \lambda y : \sigma_1. u : \sigma_1 \rightarrow \sigma_2} \text{LAM}$$

By **weakening** we have $\Gamma, y : \sigma_1 \vdash e : \tau$. Then, applying the **IH** to that and the smaller derivation $\Gamma, x : \tau, y : \sigma_1 \vdash u : \sigma_2$ we obtain a derivation of $\Gamma, y : \sigma_1 \vdash u[e/x] : \sigma_2$. We can then construct

$$\frac{\displaystyle \frac{\vdots}{\Gamma, y : \sigma_1 \vdash u[e/x] : \sigma_2}}{\Gamma \vdash \lambda y : \sigma_1. u[e/x] : \sigma_1 \rightarrow \sigma_2} \text{LAM}$$

But we have that $(\lambda y : \sigma_1. u)[e/x] \stackrel{\text{def}}{=} \lambda y : \sigma_1. u[e/x]$, so we have the result.

Case(APP). Suppose the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma, x : \tau \vdash u_1 : \tau_2 \rightarrow \sigma} \quad \frac{\vdots}{\Gamma, x : \tau \vdash u_2 : \tau_2}}{\Gamma, x : \tau \vdash u_1(u_2) : \sigma} \text{ APP}$$

Applying the **IH** to the smaller derivation $\Gamma, x : \tau \vdash u_1 : \tau_2 \rightarrow \sigma$ we obtain $\Gamma \vdash u_1[e/x] : \tau_2 \rightarrow \sigma$.

Applying the **IH** to the smaller derivation $\Gamma, x : \tau \vdash u_2 : \tau_2$ we obtain $\Gamma \vdash u_2[e/x] : \tau_2$.

We combine these two derivations using the rule APP:

$$\frac{\frac{\vdots}{\Gamma \vdash u_1[e/x] : \tau_2} \quad \frac{\vdots}{\Gamma \vdash u_2[e/x] : \tau_2 \rightarrow \sigma}}{\Gamma \vdash u_1[e/x](u_2[e/x]) : \sigma} \text{ APP}$$

But the subject of this last derivation is by definition $(u_1(u_2))[e/x]$. □

Claim 3 (Preservation). If $\vdash e : \tau$ and $e \mapsto e'$ then $\vdash e' : \tau$.

Proof. By induction on $e \mapsto e'$.

Case(D-BETA). Suppose the reduction is of the form $(\lambda x : \sigma. e_1)(e_2) \mapsto e_1[e_2/x]$.

We know from the assumptions that $\vdash (\lambda x : \sigma. e_1)(e_2) : \tau$. By **inversion** this means that there exists τ_2 such that $\vdash \lambda x : \sigma. e_1 : \tau_2 \rightarrow \tau$ and $\vdash e_2 : \tau_2$.

Again by **inversion** on the judgement $\vdash \lambda x : \sigma. e_1 : \tau_2 \rightarrow \tau$ we see that $\tau_2 = \sigma$, and it must be that $x : \sigma \vdash e_1 : \tau$.

By **substitution** on the judgements $x : \sigma \vdash e_1 : \tau$ and $\vdash e_2 : \tau_2$ (recall that $\sigma = \tau_2$) we get $\vdash e_1[e_2/x] : \tau$, which is what we wanted to prove.

Case(D-APP-1). Suppose that the reduction is the form $e_1(e_2) \mapsto e'_1(e_2)$ with premise $e_1 \mapsto e'_1$.

We know from the assumptions that $\vdash e_1(e_2) : \tau$. By **inversion** it must be that there exists τ_2 such that $\vdash e_1 : \tau_2 \rightarrow \tau$ and $\vdash e_2 : \tau_2$.

By the **IH** applied to $\vdash e_1 : \tau_2 \rightarrow \tau$ and $e_1 \mapsto e'_1$ we get that $\vdash e'_1 : \tau_2 \rightarrow \tau$ as well. Thus, we can construct a typing derivation

$$\frac{\frac{\vdots}{\vdash e'_1 : \tau_2 \rightarrow \tau} \quad \frac{\vdots}{\vdash e_2 : \tau_2}}{\vdash e'_1(e_2) : \tau} \text{ APP}$$

□

Claim 4 (Canonical Forms). If $\vdash e : \sigma \rightarrow \tau$ and e val, then $e = \lambda x : \sigma. u$ with $x : \sigma \vdash u : \tau$.

The proof of the canonical forms lemma is by inspection. It is a necessary lemma for proving

Claim 5 (Progress). If $\vdash e : \tau$ then either e val or $e \mapsto e'$ for some e' .

Proof. By induction on $\vdash e : \tau$.

Case(LAM). If the typing derivation ends with $\vdash \lambda x : \sigma. e : \tau$ then by the rule VAL-LAM we immediately know that $\lambda x : \sigma. e$ val, so we have the result.

Case(APP). Suppose the typing derivation is of the form

$$\frac{\frac{\vdots}{\vdash e_1 : \tau_2 \rightarrow \tau} \quad \frac{\vdots}{\vdash e_2 : \tau_2}}{\vdash e_1(e_2) : \tau} \text{ APP}$$

for some τ_2 . Then we apply the **IH** to $\vdash e_1 : \tau_2 \rightarrow \tau$, which gives two cases.

- If e_1 val then we know that $\vdash e_1 : \tau_2 \rightarrow \tau$ so by the **canonical forms** lemma we see that $e_1 = \lambda x : \tau_2. u$ for some u . Hence we can perform the reduction $(\lambda x : \tau_2. u)(e_2) \mapsto u[e_2/x]$ by the rule D-BETA.
- If there exists e'_1 with $e_1 \mapsto e'_1$ then by the rule D-APP-1 we can perform the reduction $e_1(e_2) \mapsto e'_1(e_2)$.

In either case there is an available reduction we can perform on the term $e_1(e_2)$.

□

2. (*) Do you need to prove progress and preservation for the CBV STLC, or can you say that they have already been established? Do not do any proofs in the answer to this question (unless you want to).

Solution: The dynamics of the language have changed, so there must be at least some small adjustments to the previous proofs.

Take preservation, which is proved by induction on $\mapsto$. It must certainly be proved for the new rules D-INL, D-INR, and D-APP-2. All these cases follow previous patterns, and are easy to show. It is debatable whether it needs to be shown for the rules that have changed, as these just limit previous reductions to make sure that substitutions happen only for values, and values are a subset of pre-terms—so the previous proofs essentially apply.

In the case of progress, things change a little more. Progress is the statement that any well-typed term is either a value, or a transition out of it can be taken by the dynamics. The typing rules have not changed, so the shape of the inductive proof is more or less the same. However, the available transitions have, and this might change some of the inductive cases, e.g. the one for terms of the form $e_1(e_2)$.